

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (CL) Examination
2014

CL305 Environmental Engineering-II

Date: 13th May, 2014

Day: Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What are different levels of treatment given to domestic sewage in a conventional sewage treatment plant? [03]
- (b) Explain environmental impact of waste generation. [04]
- Q - 2 (a) Differentiate between domestic and industrial wastewater. [03]
- (b) Draw a neat graph of BOD v/s incubation period. Derive a first-order equation for BOD remaining at any time t . [04]
- (c) Explain with neat sketch biological process occurring in a trickling filter. [03]
- (d) Explain briefly various methods of aeration employed in activated sludge treatment. [04]

OR

- Q - 2 (a) Differentiate between attached and suspended growth processes. [03]
- (b) Briefly describe various limitations of BOD test. [04]
- (c) How wastewater discharge is estimated? [03]
- (d) The BOD of a sewage sample incubated for seven days at 20 °C (BOD_7^{20}) has been found to be 150 mg/L. What will be the 3-day 30 °C BOD? Assume $k = 0.23 \text{ day}^{-1}$ (base 10) at 20 °C. [04]
- Q - 3 Attempt any Two: [14]
- (a) Design a grit chamber having rectangular cross section for following data:
(i) Maximum sewage flow = 25 MLD (ii) diameter of smallest grit particle to be removed = 0.2 mm (iii) average temperature = 20 °C, (iv) Sp. Gr. of grit particles = 2.5.
- (b) Design a circular primary sedimentation tank for a sewage flow of 5.4 MLD. The detention period is 2.5 h and surface overflow rate is $40 \text{ m}^3/\text{m}^2/\text{d}$.
- (c) Write short note :
1. Self-cleansing & Limiting velocities in sewers 2. Aerated grit chamber

SECTION – II

- Q - 4 (a) Define air pollution. Enlist causes of air pollution. [03]
(b) What is Environmental Audit? State importance of Environmental Audit. [04]
- Q - 5 (a) Classify air pollutants into different categories. [03]
(b) What is smog? What are the causes & effects of Smog on human health? [04]
(c) What are the general applications of the following air pollution control equipments? [03]
(i) Cyclones (ii) Fabric Filters (iii) Electrostatic precipitators
(d) Describe settling chamber with neat sketch. [04]

OR

- Q - 5 (a) Describe the effects of Carbon Monoxide & Sulfur Dioxide on human health [03]
(b) Explain various kinds of injury to plants due to air pollutants. [04]
(c) Distinguish between Dry & Wet Precipitators. [03]
(d) Explain the mechanism of action of air pollutants on human beings. [04]
- Q - 6 Attempt any Two. [14]
- (a) Explain the impacts of improper disposal of solid wastes on human health & environment in general. List various methods used for disposal of refuse.
(b) Write a note on incineration of refuse. What are its Advantages & Disadvantages?
(c) Enlist methods for indoor air pollution control.
